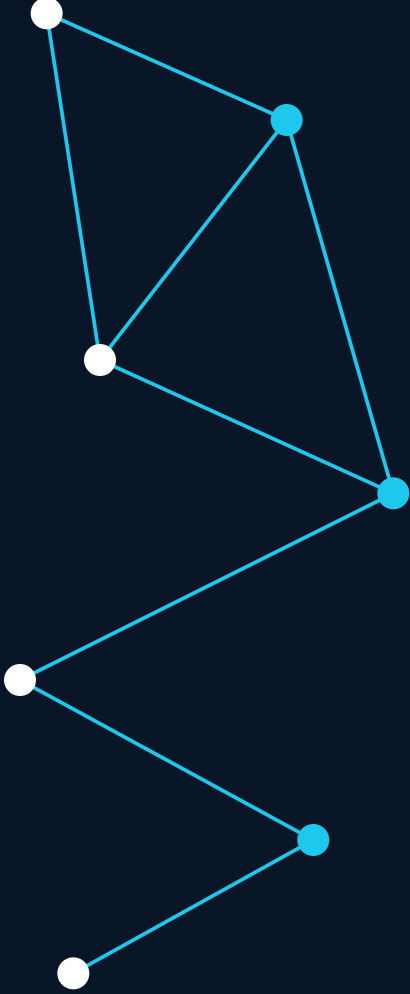


MI300X Enterprise Document RAG — boxed template conversion

DOCUMENT TITLE	AMD-ITRI Project Review: MI300X Enterprise Document RAG (boxed / peti format)
TEMPLATE FOLLOWED	AMD-ITRI PROJECT REVIEW boxed slide template (widescreen navy + cyan boxes)
VERSION	1.0 · Finalized 2026-09-11
CLASSIFICATION	Project review · Technical report · Not an SEC MD&A filing
AUTHOR (SOURCE)	Dr. Ammar Amjad · National Yang Ming Chiao Tung University (NYCU)
PLATFORM	AMD Instinct MI300X · PyTorch 2.3 · ROCm 6.2
PREPARED FROM	AMD-ITRI-Project-Review-TEMPLATE.pdf + AMD-ITRI-MI300X-Final-Comparison-Report-SOURCE.pdf
CONVERSION	All transferable text, metrics, and section boxes mapped onto the template. Source diagrams embedded.
ATTESTATION	Prepared/generated as a document-control conversion. Not a handwritten, cryptographic, or legal personal signature.
PAGE COUNT	14 pages including this control sheet



AMD – ITRI PROJECT REVIEW

MI300X Enterprise Document RAG Project Report

Enterprise document retrieval and source-grounded Chinese answer generation

Dr. Ammar Amjad

National Yang Ming Chiao Tung University

AMD Instinct MI300X • PyTorch 2.3 / ROCm 6.2 • Implemented RAG prototype

Also titled in the comparison source: AMD-ITRI High-Performance Computing Platform
Boxed conversion of the Final Comparison Report onto this Project Review template

RAG turns scattered PDF and Excel content into grounded answers

BUSINESS PROBLEM

Enterprise knowledge is spread across PDF reports and Excel sheets. Finding the right information manually is slow, and answers often lack a traceable source.

IMPLEMENTED SOLUTION

Retrieval-Augmented Generation (RAG): the system first retrieves the most relevant document chunks, then Qwen2.5-7B-Instruct generates a Traditional Chinese answer grounded in them.

1 Data Preparation

PDF + Excel → text extraction → 500-char chunks → BGE embeddings → vector index

2 Retrieval

User question → query embedding → dot-product similarity → Top-3 chunks

3 Generation

Retrieved context + sources → Qwen2.5-7B-Instruct → Traditional Chinese answer with filenames

Objective: a searchable knowledge base that answers with sources

PROBLEM ADDRESSED

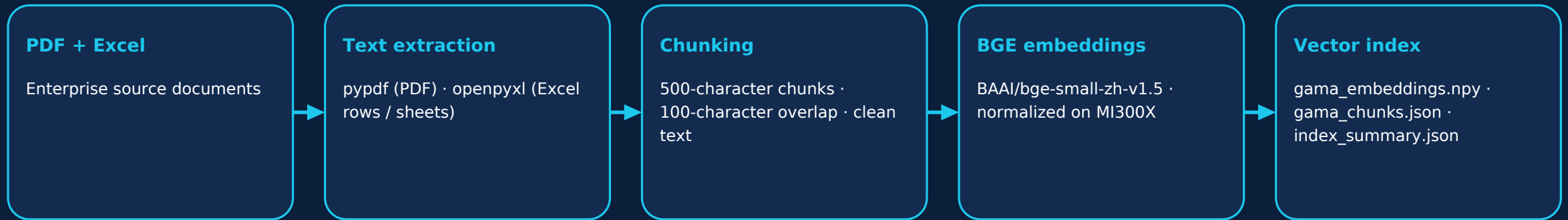
- Information distributed across PDF and Excel files
- Manual search is time-consuming
- Answers may lack document sources
- Answers need traceable document sources (comparison source wording)



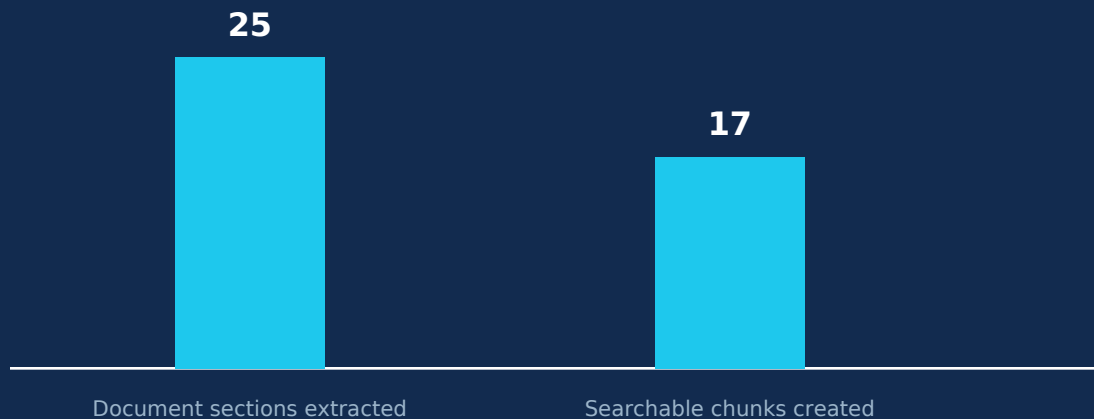
IMPLEMENTED OBJECTIVE

- Build a searchable document knowledge base / Chinese document knowledge index
- Retrieve relevant chunks for a question
- Generate source-grounded Traditional Chinese answers with Qwen

Documents become a persistent vector index in five steps



Verified index results



INDEX FACTS

- 25 document sections extracted
- 17 searchable text chunks created
- Embeddings normalized for dot-product similarity
- Index persisted to disk for reuse across queries
- Saved files: gama_embeddings.npy, gama_chunks.json, index_summary.json

The architecture links document evidence to the generated answer

OFFLINE · INDEX CONSTRUCTION



ONLINE · QUESTION ANSWERING



COMPARISON-SOURCE IMPLEMENTATION FACTS (from the architecture diagram)

- Embedding path uses BAAI/bge-small-zh-v1.5 via Sentence Transformers; batch size 32; normalized vector embeddings.
- Index stores source + page + sheet + row metadata with gama_embeddings.npy and gama_chunks.json.
- Similarity: $\text{score}(q, d) = q^T d$ on normalized vectors (cosine). Retrieval returns Top-3 relevant chunks.
- Generation: Qwen2.5-7B-Instruct, bfloat16, device_map=auto, max_new_tokens = 300, on AMD Instinct MI300X / PyTorch / ROCm.
- Observed LLM generation time recorded on the diagram: 2.55 seconds (single-query observation).

Implemented flow (source diagram placed in the template box)



Two models, two roles: retrieval and generation

BAAI/bge-small-zh-v1.5

Role: Embedding and retrieval

- Converts document chunks and user queries into semantic / comparable vectors
- Supports normalized vector search and Top-3 retrieval by dot-product similarity
- Chinese semantic embeddings (comparison source)

+

Qwen2.5-7B-Instruct

Role: Answer generation

- Uses retrieved context and system instructions
- Generates source-grounded Traditional Chinese answers
- Produces source filenames with the answer
- bfloat16 inference on MI300X

No model training or fine-tuning is claimed in this project.

Verified results on AMD Instinct MI300X

25

document sections extracted

17

searchable chunks created

2.55 s

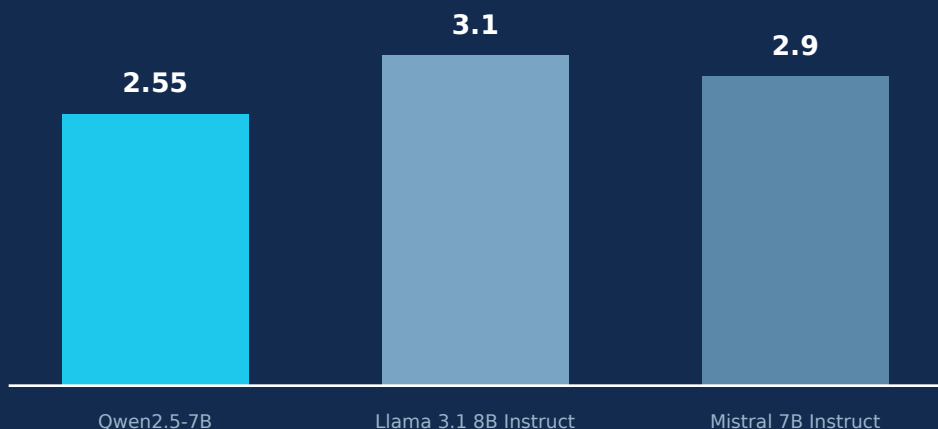
observed Qwen response-generation time

Top-3

retrieved chunks per query

Response time comparison (seconds)

From the Final Comparison Report chart — not a general GPU benchmark

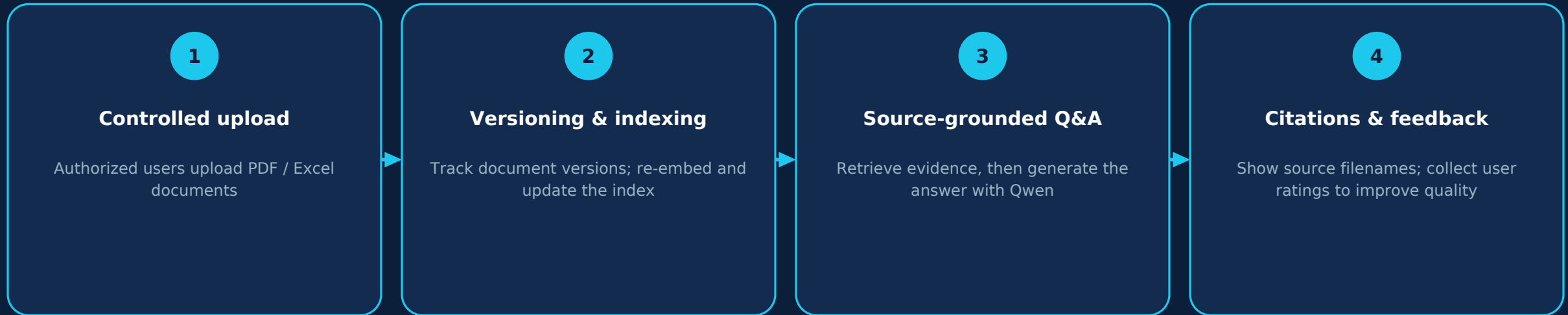


OBSERVATIONS

- Qwen2.5-7B-Instruct loaded and executed successfully on MI300X (bfloat16, device_map=auto).
- The answer used the retrieved documents and stated “insufficient information” instead of inventing a third source.
- Execution environment: AMD Instinct MI300X • PyTorch 2.3 • ROCm 6.2.
- Note: 2.55 seconds is an observed single-query generation time, not a general GPU benchmark.

Proposed deployment model for enterprise RAG

PROPOSED · NOT YET IMPLEMENTED



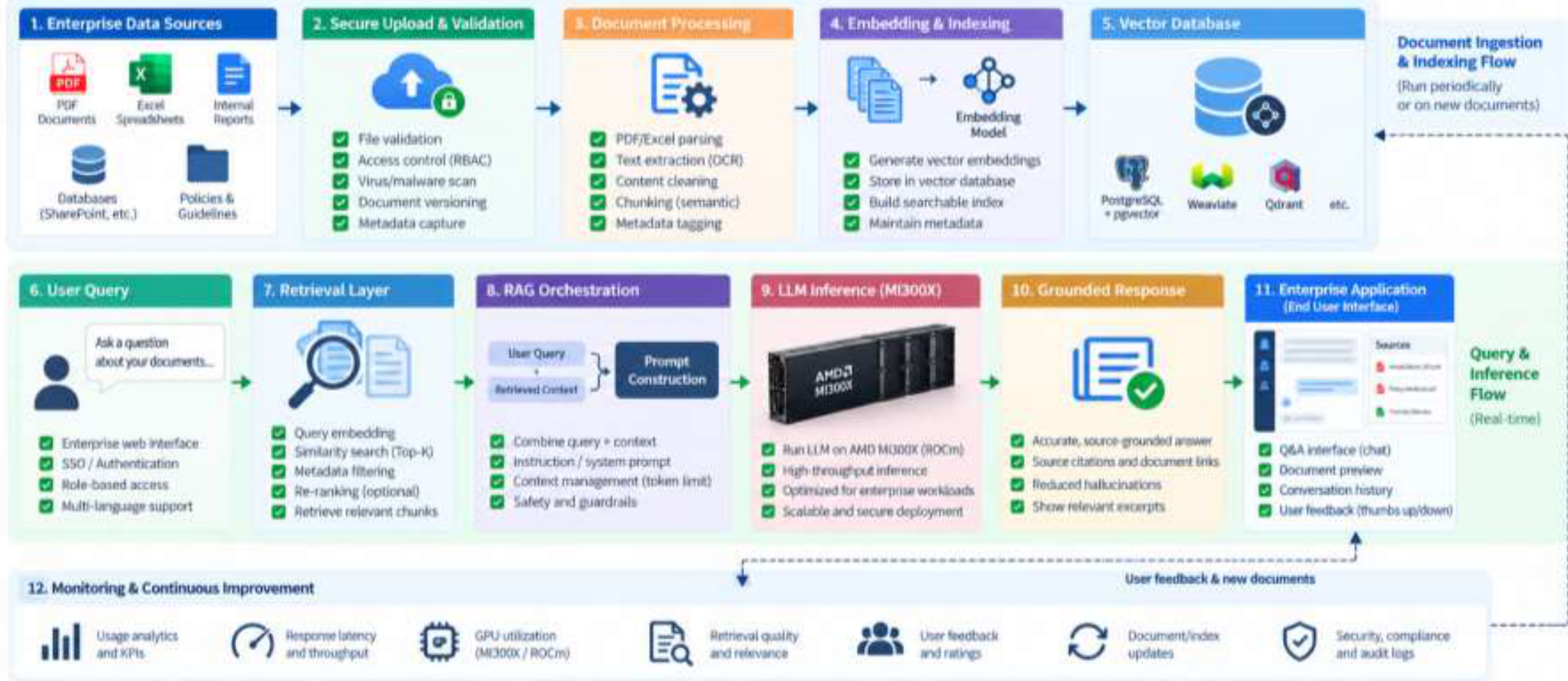
FUTURE-STATE INTERFACE (CONCEPT) · FINAL PRODUCT FUNCTIONS FROM COMPARISON SOURCE

Q: Ask a question about your enterprise documents...

A: Grounded answer · Sources: report.pdf, data.xlsx · Was this helpful?
Yes / No

Proposed enterprise flow (source diagram in the template box)

Visual from the Final Comparison Report. Labels transcribed on the next page. Proposed / not claimed as implemented.



Proposed enterprise flow — boxed transcription of the source diagram

1. ENTERPRISE DATA SOURCES

PDF documents; Excel spreadsheets; internal reports; databases (SharePoint, etc.); policies & guidelines.

2. SECURE UPLOAD & VALIDATION

File validation; access control (RBAC); virus/malware scan; document versioning; metadata capture.

3. DOCUMENT PROCESSING

PDF/Excel parsing; text extraction (OCR noted on diagram); content cleaning; semantic chunking; metadata tagging.

4. EMBEDDING & INDEXING

Generate vector embeddings; store in vector database; build searchable index; maintain metadata.

5. VECTOR DATABASE

PostgreSQL + pgvector, Weaviate, Qdrant, etc. Document ingestion periodically or on new documents.

6. USER QUERY

Enterprise web interface; SSO / authentication; role-based access; multi-language support.

7. RETRIEVAL LAYER

Query embedding; similarity search (Top-K); metadata filtering; optional re-ranking; retrieve relevant chunks.

8. RAG ORCHESTRATION

Combine query + retrieved context; instruction / system prompt; token-limit context management; safety and guardrails.

9. LLM INFERENCE (MI300X)

Run LLM on AMD MI300X (ROCm); high-throughput inference; optimized for enterprise workloads; scalable and secure deployment.

10. GROUNDED RESPONSE

Source-grounded answer; citations and document links; reduced hallucinations; show relevant excerpts.

11. ENTERPRISE APPLICATION

Q&A chat interface; document preview; conversation history; user feedback (thumbs up/down).

12. MONITORING & IMPROVEMENT

Usage analytics/KPIs; latency and throughput; GPU utilization (MI300X / ROCm); retrieval quality; ratings; index updates; security, compliance, and audit logs.

Request for renewed MI300X access

WHAT THE INITIAL ALLOCATION ACHIEVED

The first MI300X allocation validated the technical feasibility of the full workflow on AMD hardware with PyTorch 2.3 / ROCm 6.2:

- Document indexing (PDF + Excel → 17 chunks)
- Normalized Top-3 semantic retrieval
- Qwen2.5-7B-Instruct inference (bfloat16)
- Source-grounded Traditional Chinese answers

PLANNED NEXT STEPS WITH RENEWED ACCESS

- 1 Expand official product and service documents
- 2 Create a 50-100 question evaluation set
- 3 Measure retrieval quality and inference performance reproducibly / systematically
- 4 Build the proposed enterprise knowledge interface

Expected deliverables and closing

EXPECTED DELIVERABLES (comparison source)

- Reproducible MI300X benchmark report
- Validated source-grounded RAG results
- A proposed interface for enterprise knowledge support

Thank you to AMD and ITRI for enabling this work.

Prepared/generated conversion · 2026-09-11 · Document-control attestation only

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